



Agent-Based Simulation of Forest Fire Spread with NetLogo

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Abstract. Forest fires represent a significant and growing threat to natural ecosystems and human settlements, with their unpredictable behavior and capacity for rapid expansion over time, creating substantial challenges for effective prevention, control, and mitigation. This paper presents the development of a forest fire simulator designed to model and predict fire spread under varying environmental conditions. Such a simulator must consider how fire spreads in different locations and climate conditions, showing the final shape of the fire in a given period of time. Using the NetLogo agent-based modeling platform, a simulated forest environment was created in which trees function as autonomous agents interacting with one another and the environment. Identifying and understanding the risk factors that increase the likelihood of a fire occurring, as well as those that contribute to its spread and intensity, is essential for the development of an accurate forest fire simulator. Such a simulator can integrate the complex interactions among these variables to produce dynamic visualizations of fire progression, allowing users to evaluate different scenarios and make informed decisions for preventing, controlling and fighting forest fires. By incorporating key factors—such as vegetation density, temperature, humidity, topography, and wind direction—the system calculates the probability of fire propagation and generates visual representations of fire behavior over time. This tool allows users to forecast fire behavior and assess response strategies proactively, thereby improving the accuracy and efficiency of firefighting efforts. In addition, the simulator yields significant social benefits, especially for older adults residing in fire-prone areas, by supporting early warning systems, enabling prompt evacuations, and mitigating their susceptibility to fire-related risks through enhanced preparedness and coordinated response measures.

Keywords: Forest fires · Risk factors · Fire simulations

1 Introduction

A forest fire is a destructive and uncontrolled event characterized by the presence of flames, intense heat, release of smoke and the ability to spread rapidly, causing significant damage to objects, structures and, most importantly, human life. Unlike other natural disasters, fires are not limited to a specific period of time and, if not properly controlled, have a tendency to develop over time [4].

The forest fires are influenced by several conditions or elements, usually called by risk factors, that increase the likelihood of a forest fire occurring or that contribute to its spread and intensity. These factors can include natural elements such as a dry climate, high temperatures, low air humidity and strong winds, which create favorable conditions for the rapid spread of fire. Additionally, the presence of flammable fuels such as dry vegetation, dense shrubs, dense forests, and accumulated materials increases the risk of forest fires. Human factors also play a significant role as risk factors in forest fires, including negligent or intentional activities such as improperly extinguished fires and improper handling of flammable equipment or materials. Additionally, lack of awareness and education about fire prevention and safety can contribute to increased risk. Another important risk factor is the topography of the terrain, which can affect the speed of propagation and the difficulty of access to fight the forest fire. Steep areas, narrow canyons or remote locations can make it difficult for firefighters to respond and apply effective control strategies. Identifying and understanding these risk factors is essential for preventing and fighting forest fires, allowing the implementation of preventive measures and the adoption of appropriate fire-fighting strategies.

This paper addresses the development of a simulator designed to predict fire spread across various scenarios, taking into account multiple risk factors. By modeling how a fire is likely to evolve from a specific point of origin, the simulator aims to enhance decision-making processes and enable more effective firefighting strategies. Such a tool holds significant potential for improving ignition risk control, optimizing resource allocation, and minimizing the environmental and social impacts of forest fires. By visualizing the projected shape and extent of a fire over time, it can provide valuable insights for both emergency response teams and long-term planning efforts, ultimately contributing to better protection of natural ecosystems and vulnerable communities.

2 Literature Review

A review of existing fire simulation tools and related research was conducted and presented in this section. The reviewed literature highlights a range of modeling approaches, each offering unique capabilities in simulating fire spread and aiding fire management tools.

Morales et al. (2014) argue that the incorporation of simulation tools into fire prevention systems can significantly enhance operational efficiency. For instance, the Bogotá Fire Department (UAECOBB) could benefit from a simulator capable of predicting the fire's trajectory based on geographic and meteorological inputs. Such a system would assist in designing effective response strategies and resource allocation plans by displaying the projected fire perimeter over time [6].

Alessandri et al. (2021) emphasize the critical need for predictive modeling tools, given the growing threat of forest fires to human populations and protected natural areas. The authors highlight the value of full-physics models, which offer high accuracy by simulating the intricate dynamics of fire propagation taking into account many of the phenomena involved in fire propagation, together with their mutual interactions. These models consider local and regional climate conditions, terrain features, and fuel characteristics, and are also capable of modeling secondary effects such as smoke production and atmospheric dispersion [1].

Denham et al. (2022) note that forest fires are increasing in frequency and severity worldwide, posing considerable environmental and ecological risks. The causes of forest fires vary, with the vast majority (approximately 95%) linked to human activities such as land speculation, negligence, and uncontrolled burning [2].

In the initial stages of development, the forest ecosystems exhibit characteristics similar to shrublands, marked by elevated flammability. As the forests mature, their susceptibility to fire generally declines, attributed to the development of a closed canopy and the vertical stratification that separates the understory from surface fuels. The mechanisms governing ignition are highly variable and depend on fuel type, including distinctions between live and dead biomass, organic and inorganic components, as well as fine versus coarse materials, and the moisture content of those fuels. While fuel load plays a critical role in sustaining fire propagation, it is essential to categorize vegetation types based on their influence on fire dynamics. Additionally, previous weather conditions significantly affect fuel moisture and thus influence fire behavior, underscoring the importance of short-term climatic patterns in wildfire risk assessments.

During an active forest fire, precipitation can suppress and potentially extinguish the advancing fire front, whereas wind acts as a key driver of fire behavior by supplying oxygen to combustion processes, enhancing the horizontal spread of flames, and dispersing smoke, which may expedite the drying of nearby fine fuels [7,9]. Topographical features also play a significant role in fire dynamics; in uneven terrain, fire tends to propagate more rapidly upslope and more slowly downslope, primarily due to the effects of convective heat transfer and buoyancy-driven airflow [2,8].

2.1 Forest Fire Simulators

Morales (2014) presents a forest fire simulator structured in two distinct phases: data processing and simulation. In the first phase, input data, such as environmental and fire-related parameters, are classified into numerical attributes that relate propagation speeds to specific characteristics. The second phase involves

the use of these attributes to simulate the fire's spread from a point of origin, predicting propagation speed under different conditions [6].

FARSITE is another widely used fire simulation system, based on physical and mathematical modeling. It incorporates factors such as topography, weather, fuel type, and fire behavior to simulate forest fire progression over time and space. Its robust modeling capabilities enable the assessment of complex interactions between these variables. Additionally, FARSITE offers advanced features such as smoke production and dispersion modeling, facilitating evaluations of environmental and health impacts. Its user-friendly interface allows users to define fire scenarios and visualize the results, supporting informed decision-making in forest fire management [3].

BEHAVE is a fire modeling system that combines a variety of models and equations to predict fire behavior across various environmental scenarios. It takes into account parameters such as fuel type, humidity, wind speed, topography and terrain characteristics to calculate key metrics like rate of fire spread, flame length and fireline intensity. The simulator also offers capabilities to estimate the time needed to control and suppress the fire, aiding in strategic planning and decision-making during forest fire fighting efforts. With its user-friendly interface and advanced modeling capabilities, BEHAVE is a valuable tool for professionals involved in forest fire management and control [5].

3 Development

The goal of developing the forest simulator is to create a powerful and accurate tool for modeling and simulating the behavior of forest fires. Its primary purpose is to predict the spread of forest fire in different scenarios by incorporating critical risk factors such as vegetation type and density, climatic variables (e.g., temperature, humidity, and wind), and topographical features. Through the simulation of forest fire behavior, the tool seeks to improve the understanding of forest fire spread patterns, intensity, rate of expansion, and other key parameters essential for effective forest fire management. By simulating fire behavior, the simulator aims to help understand fire spread patterns, fire intensity, growth rate, and other parameters crucial to an effective forest fire response. With a robust forest simulator, it is possible to analyze and predict the evolution of the forest fire, allowing the adoption of prevention strategies, combat planning, and efficient resource allocation, contributing to the protection of ecosystems, human life, and infrastructure within forested areas.

3.1 Materials and Methods

The software used to design the simulator was NetLogo, which is an agent-based modeling and simulation platform widely used in scientific and educational research. This platform offers a user-friendly and flexible interface for creating computational models that involve the interaction between individual agents in a simulated environment (Fig. 1). Its agent-based approach allows for the

modeling of complex phenomena such as animal behavior, population dynamics, socioeconomic systems, and even ecosystems. With the NetLogo simulator, users can visualize and explore the dynamics of the model in real time, test different scenarios and hypotheses, and analyze the results obtained.

A simulated forest environment was developed, where the agents, represented by trees, are autonomous objects capable of interacting with the different elements of the environment. Patches are the individual cells of the plane that represent vegetation. The fire starts on a tree in a random position, interacts with the other elements in the scene, and the fire starts to spread.

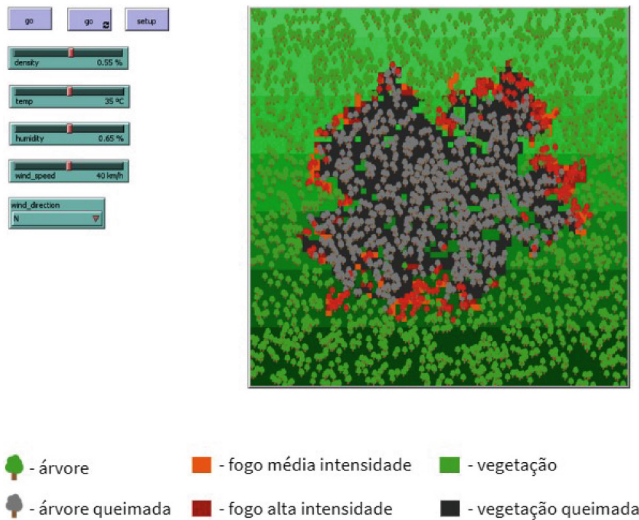


Fig. 1. Graphical Environment of the Simulator.

Once a tree agent ignites, it can transmit the fire to adjacent agents or to the patch it occupies. In turn, a burning patch can ignite the fire to the neighboring patch or to the agent that is in its position, as long as its not already on fire. To carry out this propagation, a model was used to calculate the probability of fire propagation, taking into account the risk factors involved in the simulation.

A generic fire propagation model was used to determine the probability of fire transmission. Although simplified for this study, the model aligns with risk factors commonly cited in existing literature. This modeling approach serves to simulate and predict the forest fire spread, incorporating variables such as fuel type, meteorological conditions, topography, and environmental moisture. These parameters enable the estimation of fire progression rate, affected area size, and flame intensity, thus offering valuable insights for fire behavior analysis.

Risk factors in the simulation are categorized into two types: general and individual. All risk factors are represented via user-adjustable sliders, with realistic minimum and maximum values (Fig. 2a-2d). Wind direction, due to its

categorical nature, is controlled through a combobox offering eight directional options: North, Northeast, East, Southeast, South, Southwest, West, and Northwest (Fig. 2e). General risk factors uniformly influence all agents and patches, and include tree density, ambient temperature, and humidity. Individual risk factors, which dynamically affect the fire propagation probability based on specific patch characteristics, are calculated during the simulation runtime. These include wind direction—contingent upon patch positioning—and slope, which is derived from patch elevation.

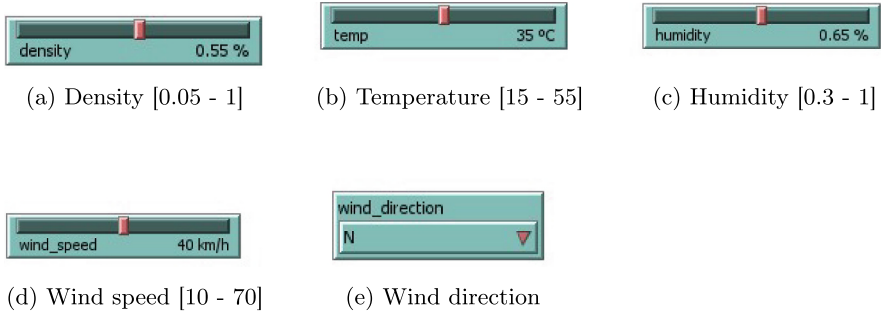


Fig. 2. Interface controls: sliders for density, temperature, humidity, wind speed, and a combo box for wind direction.

At the beginning of the simulation, the probability of forest fire spread was calculated using the general risk factors. This probability was determined according to Eq. 1, which combines the weighted influence of each factor:

$$P = w1 * T + w2 * (1 - H) + w3 * D \quad (1)$$

- P : Probability of forest fires propagation;
- T : Temperature;
- H : Humidity;
- D : Density;
- w : Coefficients.

The coefficient values for the general risk factors were determined through iterative simulations calibrated for a mild forest environment. The selected coefficients are: 0.4 for density, 0.6 for temperature, 0.8 for humidity, and 0.4 for wind speed.

The influence of individual risk factors on the computation of spread probability was incorporated within the simulation process, as these factors are contingent upon the unique attributes of each patch. Consequently, the probability must be calculated separately for each patch. For the slope variable, an initial elevation value was randomly assigned to each patch in a manner that established a general northward incline in the terrain—meaning that southern patches

were lower in elevation compared to northern ones. The slope-related probability was then derived using the Eq. 2.

$$P = P - (|dh|) \quad (2)$$

- P : Probability of forest fire propagation;
- dh : Height difference.

If $dh > 0$:

$$P = P - 0.0005$$

If $dh < 0$:

$$P = P + 0.001$$

The probability of fire spread is inversely related to the distance between patches: the greater the distance, the lower the likelihood of spread, and conversely, shorter distances are associated with higher probabilities. Furthermore, fire is more likely to spread to patches situated at higher elevations than to those at lower elevations, resulting in a faster uphill propagation compared to downhill.

To compute the probability associated with wind direction, the angle between the spatial positions of the patches is first determined and then compared to the prevailing wind direction. This procedure yields the Eq. 3.

$$P = P + \cos(|wd - a|) * w4 * ws \quad (3)$$

- P : Probability of forest fire propagation;
- wd : Wind direction;
- a : Angle formed by the Patches;
- $w4$: Wind speed coefficient;
- ws : Wind speed.

If $x2 - x1 = 0$ and $y2 >= 0$

$$a = 90$$

If $x2 - x1 = 0$ and $y2 <= 0$

$$a = 270$$

$$a = \arctan(y2 - y1/x2 - x1)$$

In this equation, the term $\cos(|wd - a|)$ yields a value ranging from -1 to 1, depending on the angular difference between the wind direction and the relative angle between patches. A difference of 0° indicates that the patch lies directly in the direction of the wind, resulting in a value of 1, whereas a difference of 180° implies that the patch is positioned directly opposite to the wind direction, yielding a value of -1. Consequently, the probability of fire spread is higher for patches aligned with the wind direction and lower for those situated against it.

4 Simulations

To test whether the simulator is faithful to reality, several simulations were carried out with constant risk factors, varying only one factor at a time using five different values for each, except for the wind direction which was tested for all possibilities. In addition, other constant factors were kept to perform better data collection and analysis, for example, the same type of vegetation with constant positions, the same origin of fire and the same slope were used. To obtain the graphs shown below (Figs. 3–13), 10 simulations were carried out for the variable risk factor and at the end the average of all the results was calculated.

These graphs represent the percentage of area burned and the peak of the fire, that is, the total amount of vegetation and trees burning simultaneously during the fire time measured in ticks.

By analyzing the graphs, we can conclude that the risk factors are acting as intended and that there are risk factors with more influence on the spread of the fire than others. For example density (Fig. 3) has a more prominent effect than wind speed (Fig. 6). We also found that increased humidity has a negative effect on the spread of fire, which is true (Fig. 4).

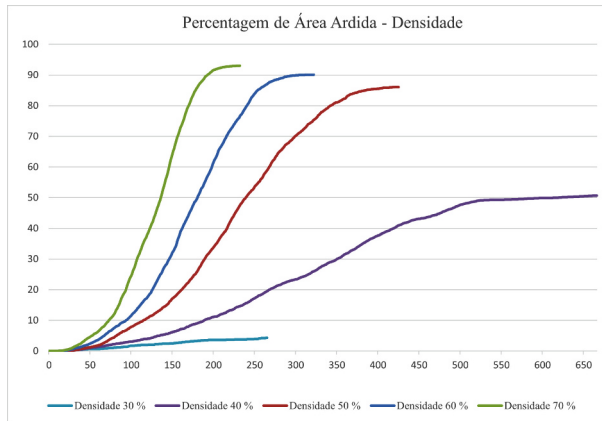


Fig. 3. Burnt percentage - Density.

The fire peak graphs show that the risk factors are acting as intended and that there are risk factors with more influence on the spread of the fire than others. For example, density (Fig. 8) has a more prominent effect than wind speed (Fig. 11), since the fire peaks faster. Also, increased humidity has a negative effect on the spread of fire, which is true (Fig. 9).

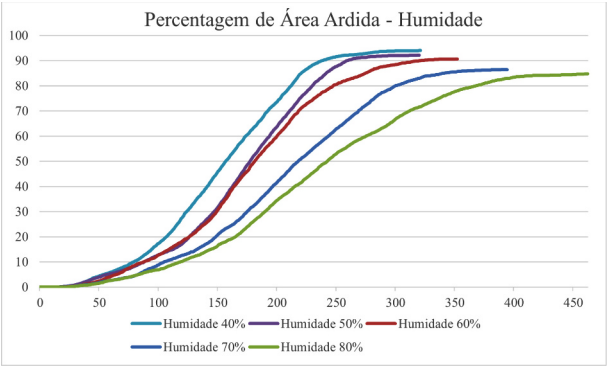


Fig. 4. Burnt percentage - Humidity.

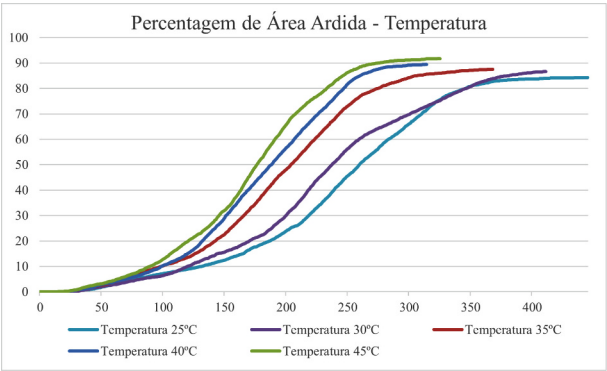


Fig. 5. Burnt percentage - Temperature.

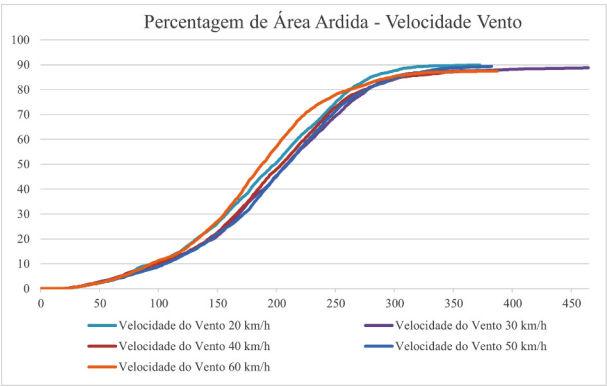


Fig. 6. Burnt percentage - Wind speed.

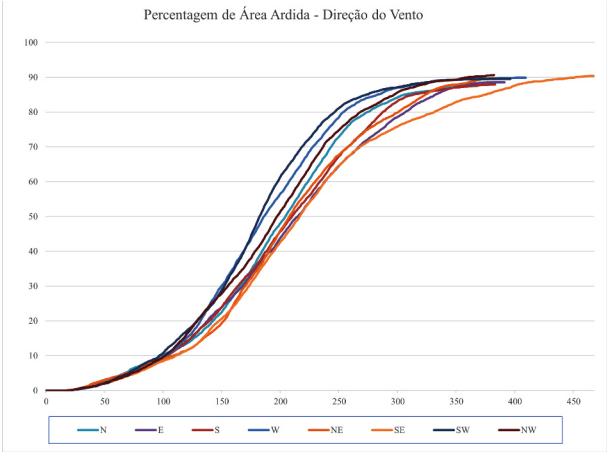


Fig. 7. Burnt percentage - Wind direction.

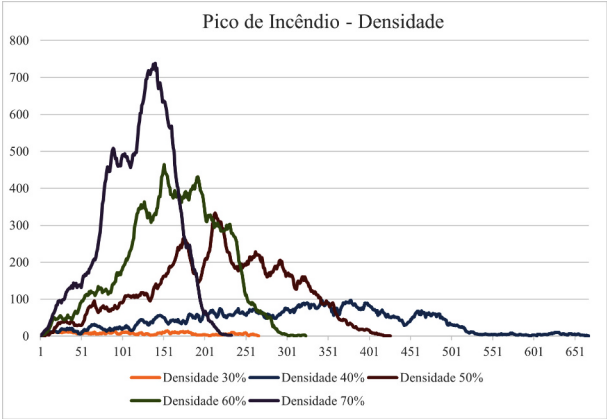


Fig. 8. Fire peak - Density.

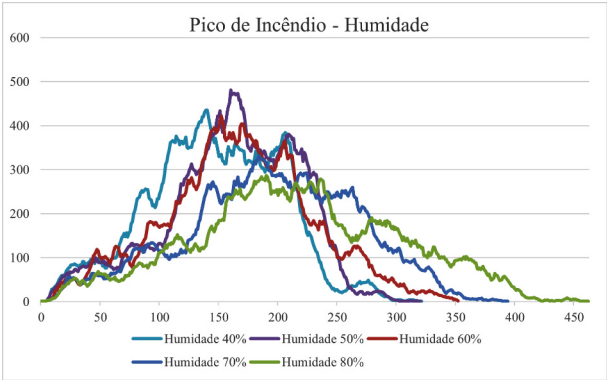


Fig. 9. Fire peak - Humidity.

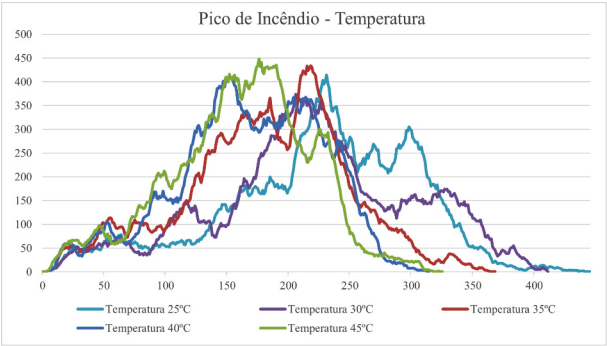


Fig. 10. Fire peak - Temperature.

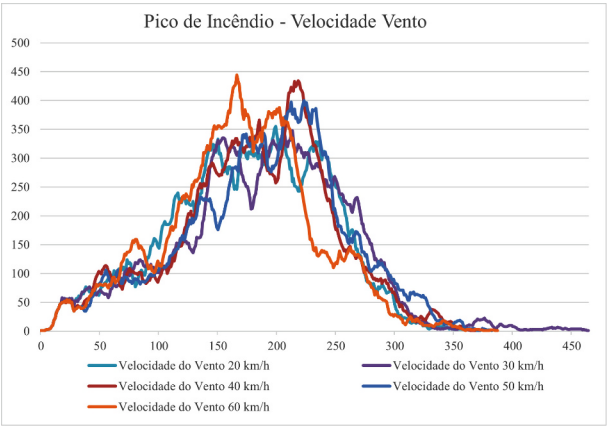


Fig. 11. Fire peak - Wind speed.

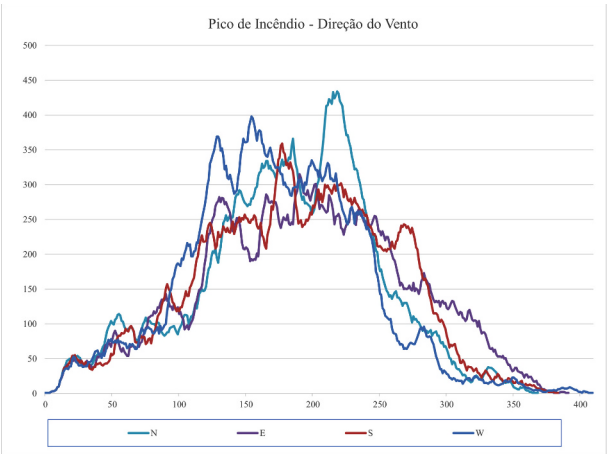


Fig. 12. Fire peak - Wind direction.

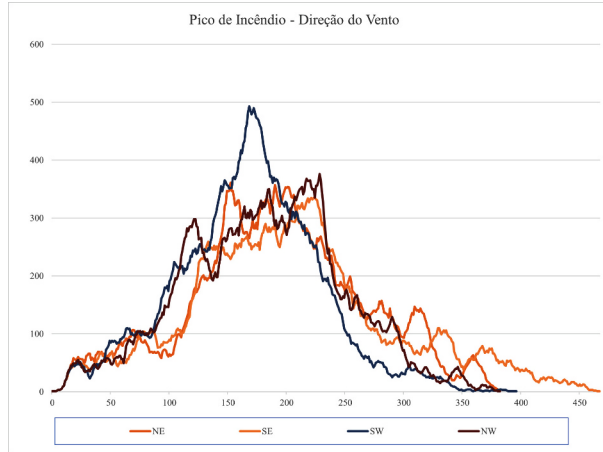


Fig. 13. Fire peak - Wind direction.

5 Conclusions and Future Work

This study focused on the development of a forest fire simulator, an essential tool for predicting and analyzing forest fire behavior. Given the severe threat forest fires pose to both natural ecosystems and human settlements, the need for effective prevention, control, and mitigation strategies is paramount. The simulator developed in this work aims to model fire spread under varying environmental scenarios by integrating multiple risk factors.

A review of established simulators, including the Morales simulator, FAR-SITE, and BEHAVE, provided valuable insights into existing modeling techniques used to simulate forest fire behaviour. These tools incorporate elements such as topography, climate, fuel types, and fire behavior to simulate fire dynamics across spatial and temporal scales. Additionally, they offer functionalities like smoke estimation and dispersion modeling, which are critical for assessing environmental and public health impacts.

A forest fire simulator was developed using the NetLogo software. The simulator creates a simulated forest environment, where trees are represented by agents that interact with the elements of the environment. Fire starts in one tree and spreads to neighboring trees and adjacent areas. A model was used to calculate the probability of fire spreading, considering factors such as vegetation type, density, weather conditions and topography.

The simulator serves as a decision-support tool, enabling better-informed strategies for fire prevention, suppression, and resource deployment. It contributes to the protection of ecosystems, human lives, and infrastructure by enhancing forest fire management capabilities. Nevertheless, it is acknowledged that the simulator represents a simplified abstraction of real-world conditions. The reliability of its predictions depends on the accuracy and granularity of input data and the fidelity of the underlying models. Ongoing refinement and

integration of updated information are essential to enhance its precision and applicability.

The development of a forest fire simulator represents a significant advancement in forest fire research and management, enabling the formulation of data-driven responses and supports the broader goal of environmental conservation and community resilience. Research in this field is crucial for addressing the complex challenges related to forest fires, aiming to protect ecosystems and ensure a sustainable future.

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